

CONCISE DESCRIPTION OF RELEVANCE: THIRD-PARTY SUBMISSION UNDER 37 CFR § 1.290

U.S. Application No. 19/318,450 (Publication No. US 2026/0070232 A1)

Document 1: U.S. Patent No. 12,570,007 (“Document 1”)

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Title: Magnetic coupling device with movable workpiece interfaces

This document discloses a magnetic coupling device with movable pole portions for magnetically coupling to a ferromagnetic workpiece, including a switchable magnetic flux source and a plurality of pole portions with workpiece interfaces. Its potential relevance to the claims of the above-identified application is set forth below.

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 2 (the displacement direction is aligned parallel to the longitudinal axis of the housing)	The reference discloses that a first movable pole portion 32 is translatable in direction 40 and direction 42 relative to a lower surface 44 of housing 14, implying a displacement direction. “In the illustrated embodiment, first movable pole portion 32 is translatable in direction 40 and direction 42 relative to a lower surface 44 of housing 14 and a lower surface 46 of north pole portion 18 .” • Paragraph [0040]
Claim 3 (The magnetic gripper according to claim 1, wherein each pole shoe has an active structure for directing the magnetic field portion)	The reference discloses that first movable pole portion 32 and workpiece engagement surface 36 are made of a ferromagnetic material to complete a magnetic circuit, thereby directing the magnetic field portion. “First movable pole portion 32 and hence workpiece engagement surface 36 are made of a ferromagnetic material to complete a magnetic circuit from switchable magnetic flux source 16 through ferromagnetic workpiece 12 .” • Paragraph [0041]
Claim 6 (The magnetic gripper according to claim 5, wherein the plurality of projections include at least a first projection and a second projection)	The reference discloses that the first plurality of projections includes a first projection, a second projection, and a third projection. “In still another example thereof, the first plurality of projections includes a first projection, a second projection, and a third projection.” • Paragraph [0018] • Paragraph [0070]

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 8 (The magnetic gripper according to claim 1, wherein the number of pole shoes includes a first pole shoe and a second pole shoe)	The abstract discloses that the plurality of pole portions includes a first pole portion including a first workpiece interface and a second pole portion including a second workpiece engagement surface. “The plurality of pole portions including a first pole portion including a first workpiece interface having a first workpiece engagement surface and a second pole portion including a second workpiece engagement surface.” • Abstract • Paragraph [0009]
Claim 9 (The magnetic gripper according to claim 1, wherein the magnet is displaceable electrically, pneumatically, or mechanically between the gripping position and the release position)	The reference discloses an actuator 84 coupled to a rare earth permanent magnet 82 to rotate it, which is a mechanical means of displacing the magnet to switch its state. “As shown in FIG. 4, an actuator 84 is coupled to rare earth permanent magnet 82 to rotate rare earth permanent magnet 82 relative to rare earth permanent magnet 80 about an axis 86 (see FIGS. 5 - 7).” • Paragraph [0045]
Claim 10 (The magnetic gripper according to claim 1, wherein the magnetic gripper has a position sensor for detecting the gripping position and/or the release position)	The reference discloses that a characteristic can be a position of one or more of the plurality of movable pole portions, which can be used to detect the gripping and/or release position. “In a variation thereof, the characteristic is a position of one or more of the plurality of movable pole portions.” • Paragraph [0010]
Claim 12 (The magnetic gripper according to claim 1, wherein the magnetic gripper has a manipulator interface for attaching the magnetic gripper to a manipulator)	The reference discloses a robotic system including a robotic arm having a magnetic coupling device attached to an end of the robotic arm, implying a manipulator interface. “A robotic system, including a robotic arm having a magnetic coupling device according to claim 1 attached to an end of the robotic arm.” • Claim 27

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 13 (The magnetic gripper according to claim 1, wherein a housing portion of the housing is designed to largely suppress the escape of the magnetic field of the magnet from the housing ...)	The reference discloses an OFF state wherein a magnetic circuit is formed within housing 14, which suppresses the escape of the magnetic field from the housing portion. “Switchable magnetic flux source 16 of magnetic coupling tool 10 is switchable between an OFF state wherein a magnetic circuit is formed within housing 14 and an ON state wherein a magnetic circuit is formed from switchable magnetic flux source ...” • Paragraph [0043]
Claim 14 (wherein, in a gripping state, the magnet is in the gripping position, and the ferromagnetic workpiece is pressed against each workpiece contact surface by means of a magnetic ...)	The reference discloses an ON state where a workpiece is held by switchable magnetic flux source 400 due to a completion of a magnetic circuit through the workpiece, which corresponds to a gripping state where the ferromagnetic workpiece is pressed against each workpiece contact surface by magnetic force. “In the on-state, a workpiece is held by switchable magnetic flux source 400 due to a completion of a magnetic circuit from the aligned north pole portions 450 of upper platter 412 and lower platter 414, through the workpiece, and ...” • Paragraph [0052] • Paragraph [0030]
Claim 16 (The magnetic gripper according to claim 1, wherein the receptacle is in the form of a groove)	The reference discloses exemplary interfaces 310 as elongated slots 312, which can be considered a type of groove. “The exemplary interfaces 310 shown in FIG. 26 are elongated slots 312 having lower surfaces 314 and upper surfaces 316 .” • Paragraph [0072]